

Importing & Securing Data in ServiceNow

Problem Statement:

Linking each record to an employee and pulling some employee details (like department) into the record for easier reporting.

Objective:

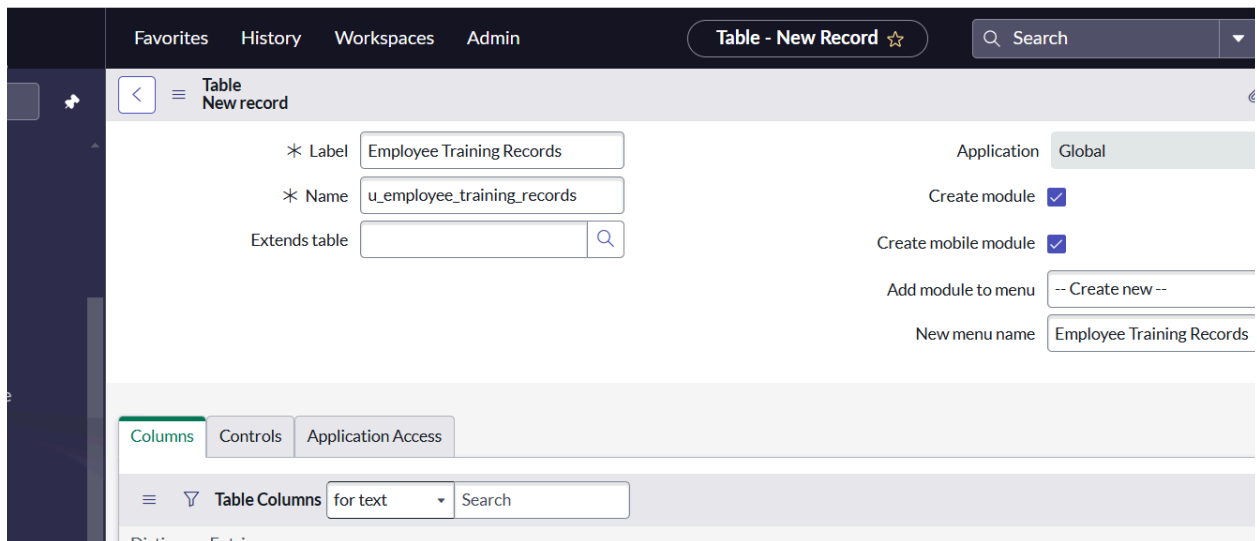
The objective of this initiative is to implement an Access control list by using Impersonate user. This provides a solid foundation for managing and securing data in servicenow using key platform features.

TASK INITIATION

Milestone 1 : Tables

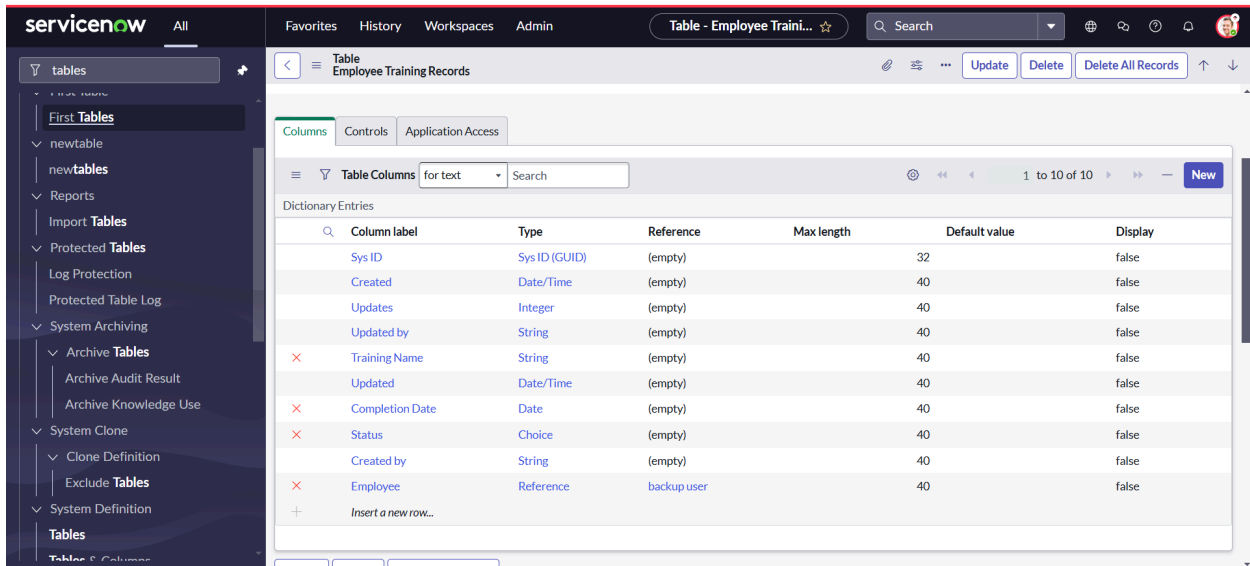
Activity 1: Create Table

1. Open service now.
2. Click on All >> search for Tables
3. Select Tables under system Definition.
4. Click on new
5. Fill the following details to create a new Table



6. Add the following fields under the columns section:

- Training Name (Type: String)
- Completion Date (Type: Date)
- Status (Type: Choice)
- Employee (Type: Reference), (Reference field to sys_user table)

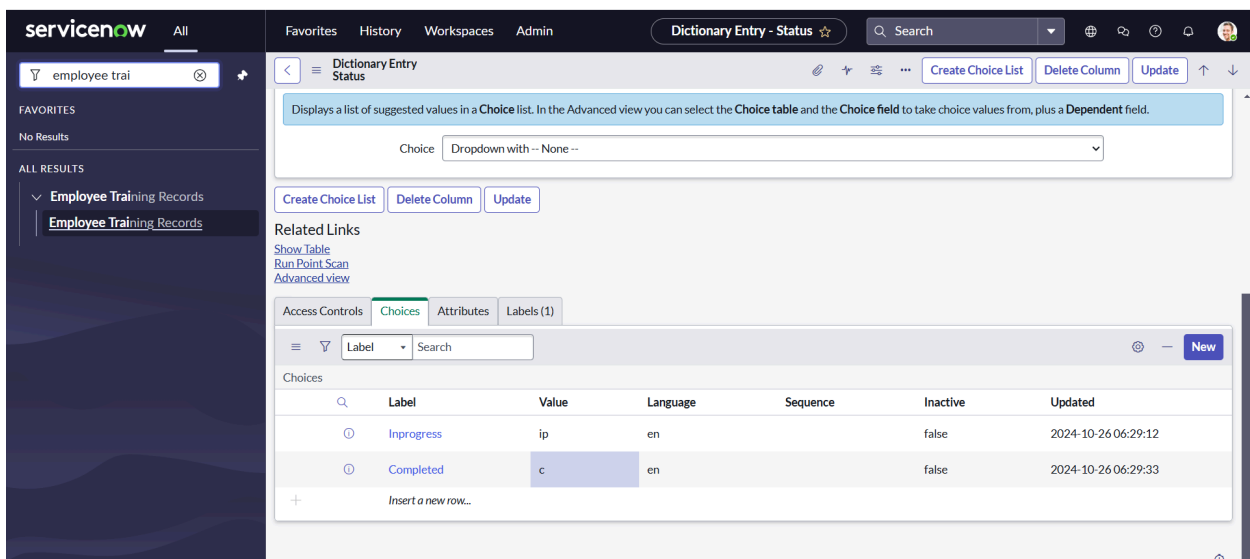


The screenshot shows the ServiceNow interface for editing the 'Employee Training Records' table. The 'Dictionary Entries' tab is active, displaying a list of columns and their properties.

Column label	Type	Reference	Max length	Default value	Display
Sys ID	Sys ID (GUID)	(empty)	32		false
Created	Date/Time	(empty)	40		false
Updates	Integer	(empty)	40		false
Updated by	String	(empty)	40		false
Training Name	String	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Completion Date	Date	(empty)	40		false
Status	Choice	(empty)	40		false
Created by	String	(empty)	40		false
Employee	Reference	backup user	40		false

7. Click on submit

8. Click on Status Field to add choices in the Dictionary Entry Status



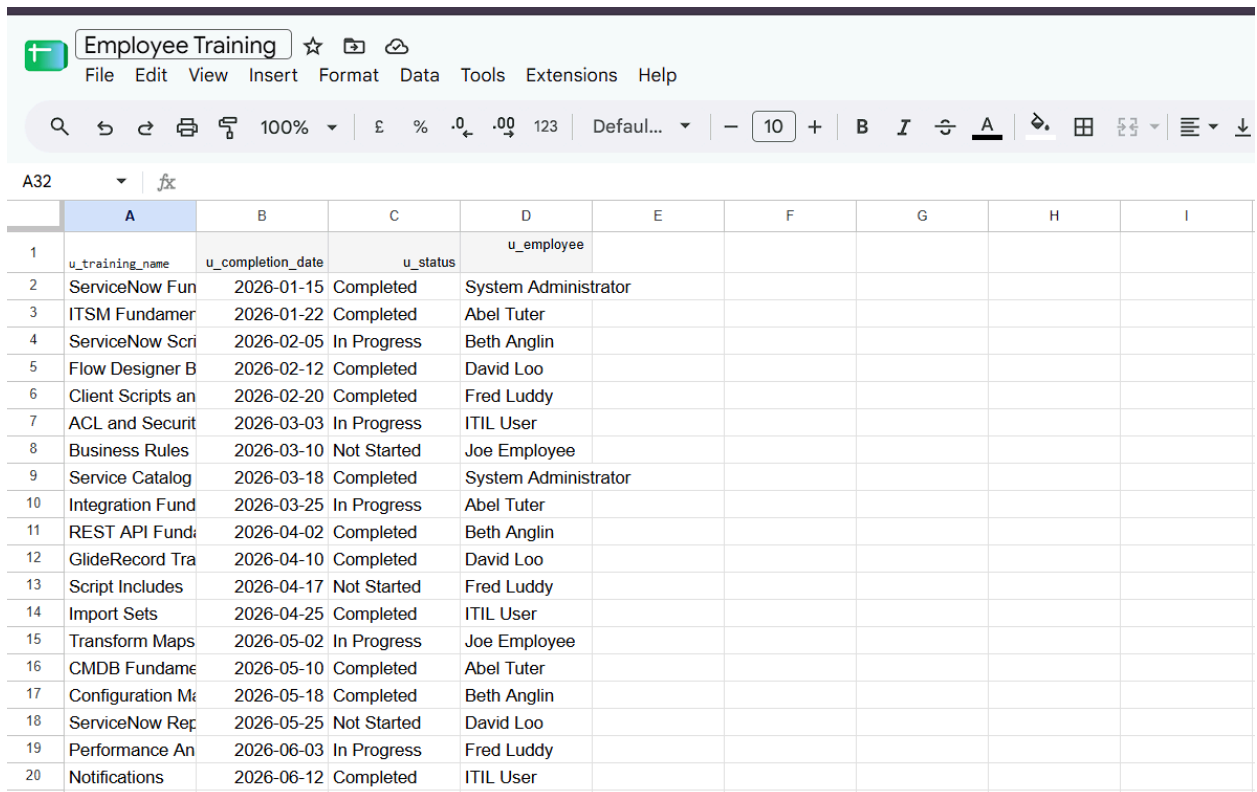
The screenshot shows the ServiceNow interface for the 'Dictionary Entry - Status' table. The 'Choices' tab is active, displaying a list of choices.

Label	Value	Language	Sequence	Inactive	Updated
Inprogress	ip	en		false	2024-10-26 06:29:12
Completed	c	en		false	2024-10-26 06:29:33

Milestone 2 : Import Data

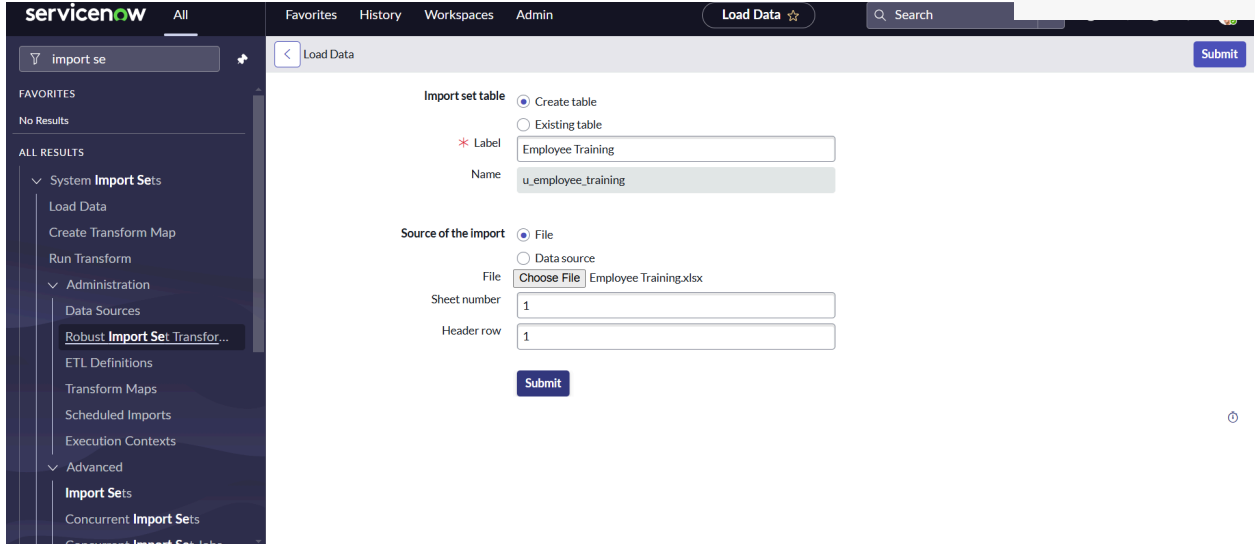
Activity 1: Importing Data

1. Create a .xlsx file using the spread sheets with the four fields (Training Name, Completion Date, Status and Employee) and download it.



	A	B	C	D	E	F	G	H	I
1	u_training_name	u_completion_date	u_status	u_employee					
2	ServiceNow Fun	2026-01-15	Completed	System Administrator					
3	ITSM Fundamer	2026-01-22	Completed	Abel Tuter					
4	ServiceNow Scri	2026-02-05	In Progress	Beth Anglin					
5	Flow Designer B	2026-02-12	Completed	David Loo					
6	Client Scripts an	2026-02-20	Completed	Fred Luddy					
7	ACL and Securit	2026-03-03	In Progress	ITIL User					
8	Business Rules	2026-03-10	Not Started	Joe Employee					
9	Service Catalog	2026-03-18	Completed	System Administrator					
10	Integration Fund	2026-03-25	In Progress	Abel Tuter					
11	REST API Funda	2026-04-02	Completed	Beth Anglin					
12	GlideRecord Tra	2026-04-10	Completed	David Loo					
13	Script Includes	2026-04-17	Not Started	Fred Luddy					
14	Import Sets	2026-04-25	Completed	ITIL User					
15	Transform Maps	2026-05-02	In Progress	Joe Employee					
16	CMDB Fundame	2026-05-10	Completed	Abel Tuter					
17	Configuration M	2026-05-18	Completed	Beth Anglin					
18	ServiceNow Rep	2026-05-25	Not Started	David Loo					
19	Performance An	2026-06-03	In Progress	Fred Luddy					
20	Notifications	2026-06-12	Completed	ITIL User					

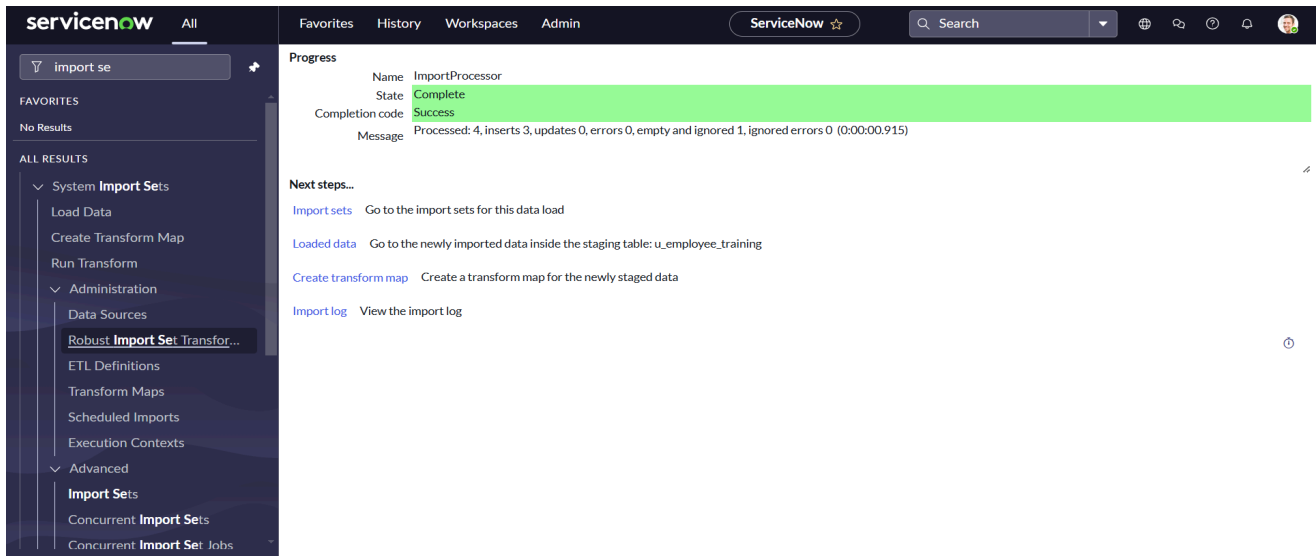
2. Open service now.
3. Click on All >> System Import Sets.
4. Select Load Data and Upload File that you have already created with four fields that are:(Training Name, Completion Date, Status and Employee)
5. Label: Employee Training
6. Name: u_employee_training



The screenshot shows the 'Load Data' form in ServiceNow. The left sidebar contains a navigation menu with 'Import Sets' highlighted. The main form area has the following fields:

- Import set table:**
 - ☒ Create table
 - ☐ Existing table
- Label:** Employee Training
- Name:** u_employee_training
- Source of the import:**
 - ☒ File
 - ☐ Data source
- File:** Choose File Employee Training.xlsx
- Sheet number:** 1
- Header row:** 1
- Submit:** A blue button at the bottom right of the form.

7. Click on submit



The screenshot shows the 'Progress' page in ServiceNow. The left sidebar contains a navigation menu with 'Import Sets' highlighted. The main form area has the following sections:

- Progress:**
 - Name:** ImportProcessor
 - State:** Complete
 - Completion code:** Success
 - Message:** Processed: 4, inserts 3, updates 0, errors 0, empty and ignored 1, ignored errors 0 (0:00:00.915)
- Next steps...**
 - [Import sets](#) Go to the import sets for this data load
 - [Loaded data](#) Go to the newly imported data inside the staging table: u_employee_training
 - [Create transform map](#) Create a transform map for the newly staged data
 - [Import log](#) View the import log

Activity 2: Map Fields

1. Open Service Now
2. Click on All >> System Import Sets>> Administration >> Transform Maps
3. Fill the following details to create a new Table

servicenow All Favorites History Workspaces Admin Table Transform Map - ... Search

Table Transform Map New record

* Name: Employee Transform Application: Global

* Source table: Employee Training [u_employ... Created:

Active: ☒ * Target table: Employee Training Records [u... Order: 100

Run business rules: ☒ Run script: ☐

Enforce mandatory fields: No

Copy empty fields: ☐

Create new record on empty coalesce fields: ☐

Submit

4. Click on Submit.

5. Add Field Maps as Shown

servicenow All Favorites History Workspaces Admin Table Transform Map - ... Search

Table Transform Map First Employee Update Copy Delete

Related Links

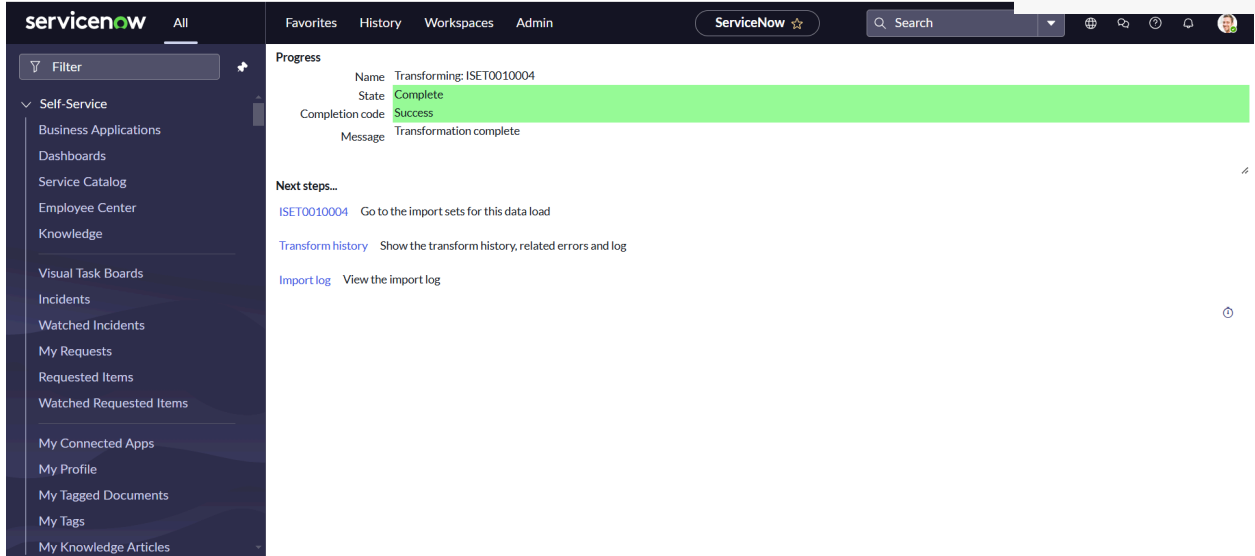
- [Auto Map Matching Fields](#)
- [Mapping Assist](#)
- [Validate Coalesce Fields](#)
- [Transform](#)
- [Index Coalesce Fields](#)
- [Run Point Scan](#)

Field Maps (4) Transform Scripts

Source field	Target field	Coalesce
u_completion_date	u_completion_date	false
u_status	u_status	false
u_training_name	u_training_name	false
u_employee_email	u_employee	false

1 to 4 of 4

6. Under Related Links Click Transform to run the import.

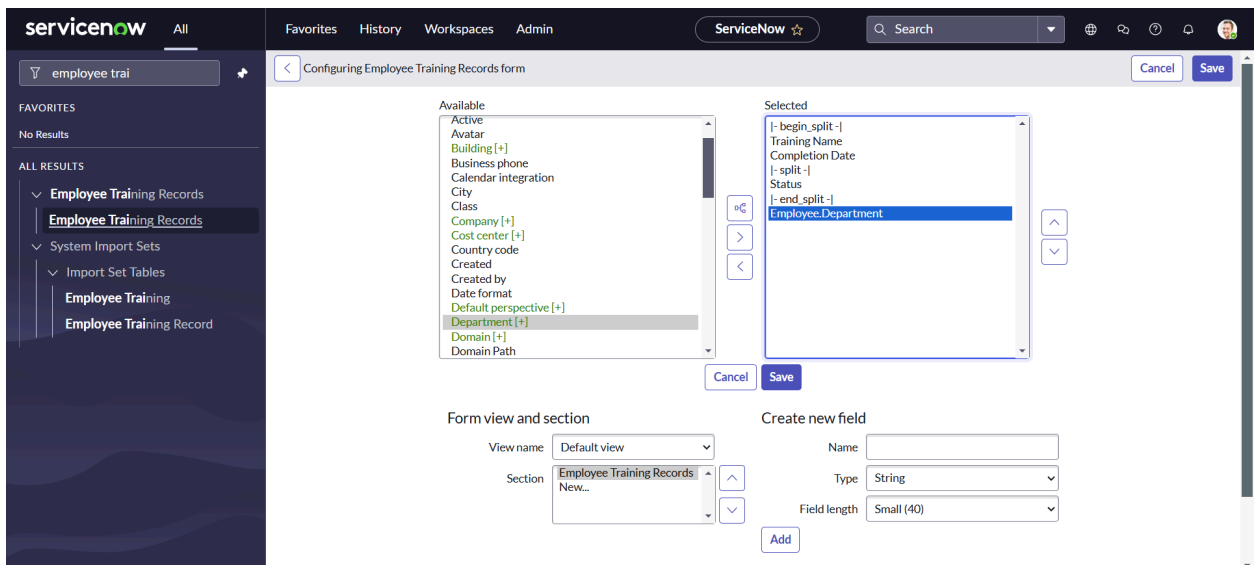


The screenshot shows the ServiceNow interface. On the left is a navigation menu with options like Self-Service, Business Applications, Dashboards, Service Catalog, Employee Center, Knowledge, Visual Task Boards, Incidents, Watched Incidents, My Requests, Requested Items, Watched Requested Items, My Connected Apps, My Profile, My Tagged Documents, My Tags, and My Knowledge Articles. The main area displays a 'Progress' section for a transformation named 'Transforming: ISET0010004'. The status is 'Complete' and the completion code is 'Success'. A message indicates 'Transformation complete'. Below this, 'Next steps...' are listed: 'Go to the import sets for this data load', 'Show the transform history, related errors and log', and 'View the import log'.

Milestone 3: Using Dot-Walking to access Employee Department Information

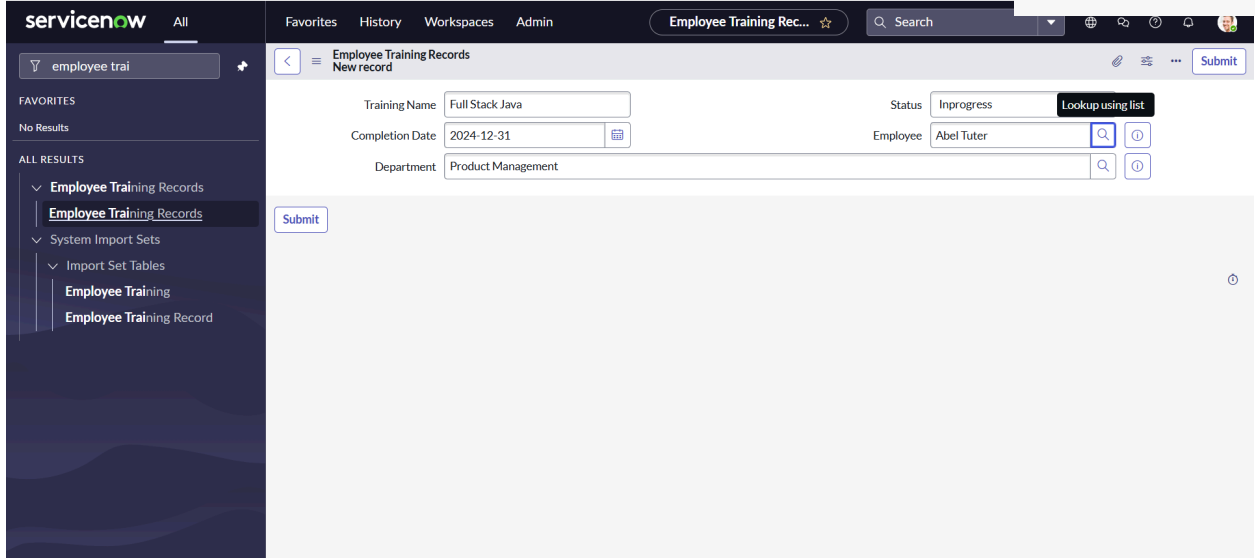
Activity 1: Dot-Walking

1. Open service now.
2. Click on All >> Employee Training Records >> Employee Training Records.
3. Open the form by clicking the New button.
4. Click the Additional Actions on the left corner and select Configure and Form Layout.
5. Click Employee[+] and dot walk and select the Department and move to the right.
6. Save.



The screenshot shows the 'Configuring Employee Training Records form' interface. On the left, a search bar contains 'employee trai'. Below it, the 'ALL RESULTS' section shows a list of items under 'Employee Training Records', including 'Employee Training Records', 'System Import Sets', 'Import Set Tables', 'Employee Training', and 'Employee Training Record'. The main area is divided into two panes: 'Available' and 'Selected'. The 'Available' pane lists various fields like Avatar, Building [+], Business phone, Calendar integration, City, Class, Company [+], Cost center [+], Country code, Created, Created by, Date format, Default perspective [+], Department [+], Domain [+], and Domain Path. The 'Selected' pane shows a list of fields including 'Employee.Department'. Below these panes, there are 'Form view and section' and 'Create new field' sections. The 'Form view and section' section has a 'View name' dropdown set to 'Default view' and a 'Section' dropdown set to 'Employee Training Records'. The 'Create new field' section has a 'Name' field, a 'Type' dropdown set to 'String', and a 'Field length' dropdown set to 'Small (40)'. There are 'Cancel' and 'Save' buttons at the bottom of the main area.

7. Now you can see the field in the List view.



The screenshot shows the ServiceNow interface for creating a new Employee Training Record. The left sidebar contains a search bar with 'employee trai' and a list of favorites including 'Employee Training Records'. The main form has the following fields:

- Training Name: Full Stack Java
- Status: Inprogress (with a 'Lookup using list' button)
- Completion Date: 2024-12-31 (with a calendar icon)
- Employee: Abel Tutor (with a search icon and a dropdown icon)
- Department: Product Management (with a search icon and a dropdown icon)

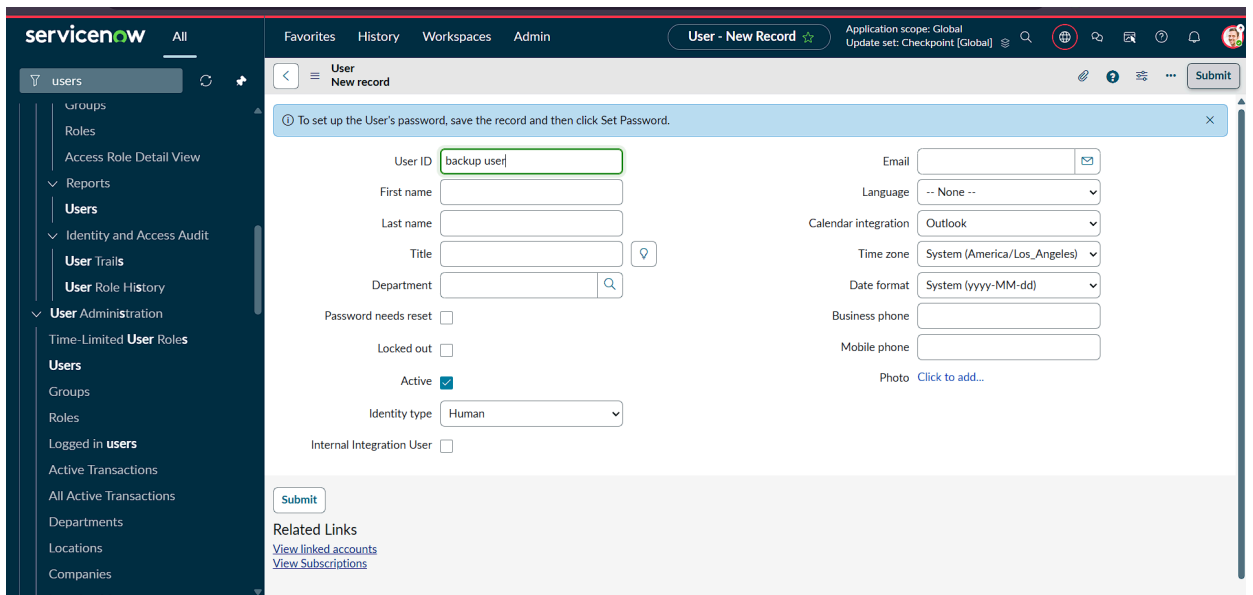
A 'Submit' button is located at the bottom left of the form.

1.

Milestone 4: Roles and User Creation

Activity 1: Create User

1. Open ServiceNow
2. Click on All>>User Administration>>Users create a new user : backup user



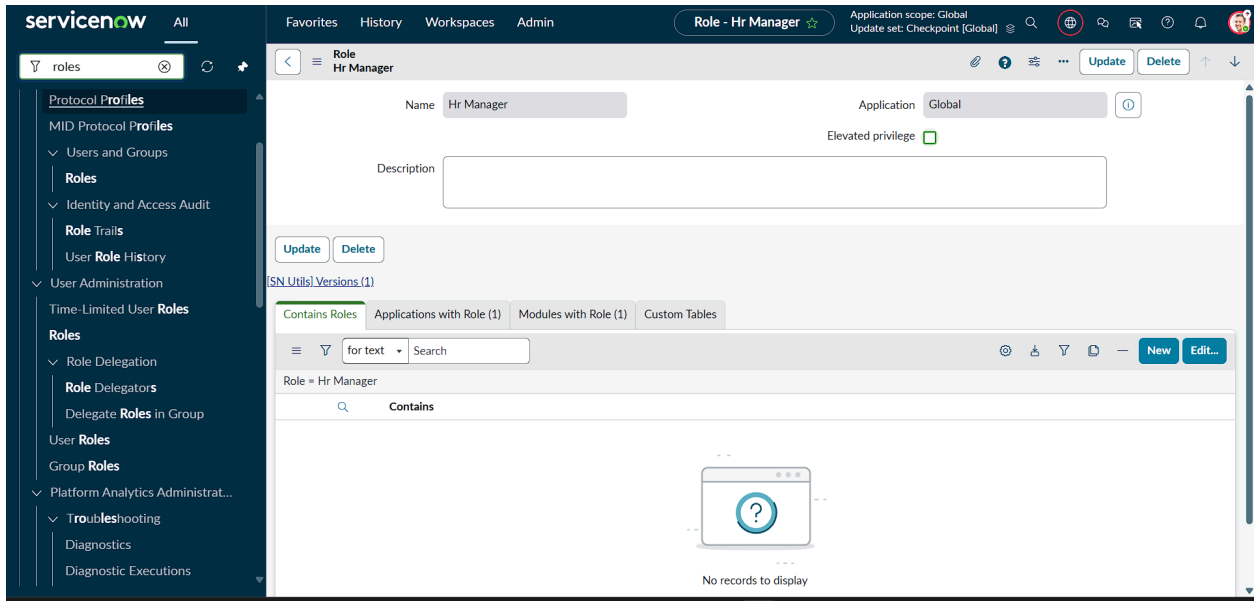
The screenshot shows the ServiceNow interface for creating a new User. The left sidebar contains a search bar with 'users' and a list of favorites including 'Users'. The main form has the following fields:

- User ID: backup user (highlighted with a green box)
- First name: (empty)
- Last name: (empty)
- Title: (empty) (with a dropdown icon)
- Department: (empty) (with a search icon)
- Password needs reset: ☐
- Locked out: ☐
- Active: ☒
- Identity type: Human (dropdown)
- Internal Integration User: ☐
- Email: (empty) (with an email icon)
- Language: -- None -- (dropdown)
- Calendar integration: Outlook (dropdown)
- Time zone: System (America/Los_Angeles) (dropdown)
- Date format: System (yyyy-MM-dd) (dropdown)
- Business phone: (empty)
- Mobile phone: (empty)
- Photo: Click to add...

A 'Submit' button is located at the bottom left of the form. A related links section at the bottom includes 'View linked accounts' and 'View Subscriptions'.

Activity 2: Create Role

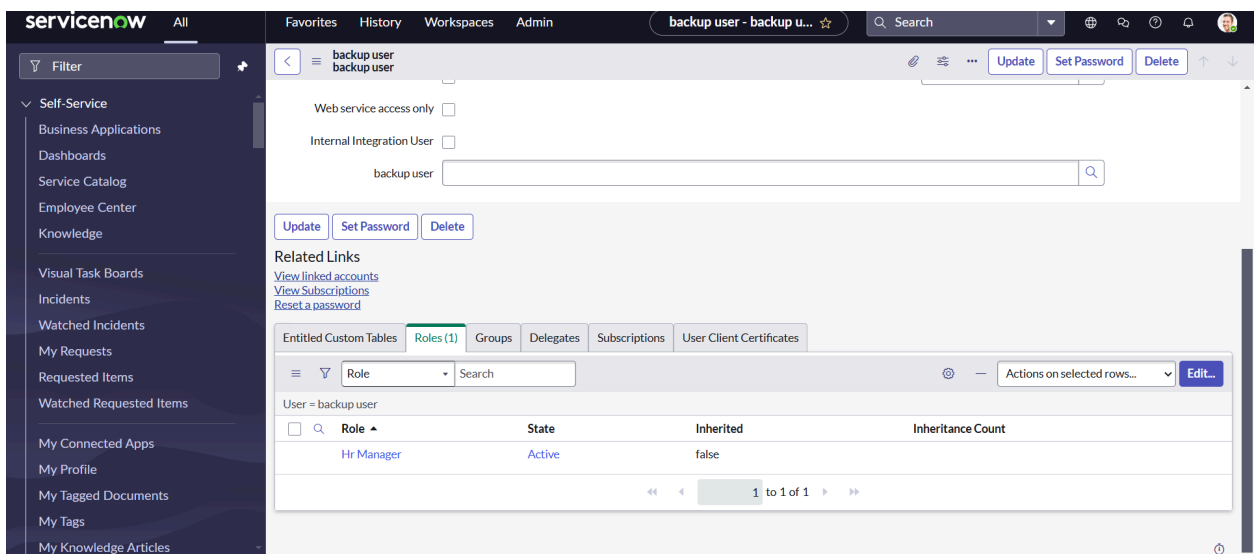
1. Open ServiceNow
2. Click on All>>User Administration>>Roles>> create a new role : Hr Manager



The screenshot shows the ServiceNow interface for creating a new role named "Hr Manager". The left sidebar contains the navigation menu with "Roles" selected under "User Administration". The main form includes fields for "Name" (Hr Manager), "Application" (Global), and "Description". Below the form are "Update" and "Delete" buttons. A section titled "Contains Roles" shows a table with no records, indicated by a "No records to display" message.

Activity 3: Assign the Role to User

1. Open ServiceNow
2. Click All>>Users (under User Definition) >>open the backup user.
3. Under roles click Edit and add the HR Manager Role.

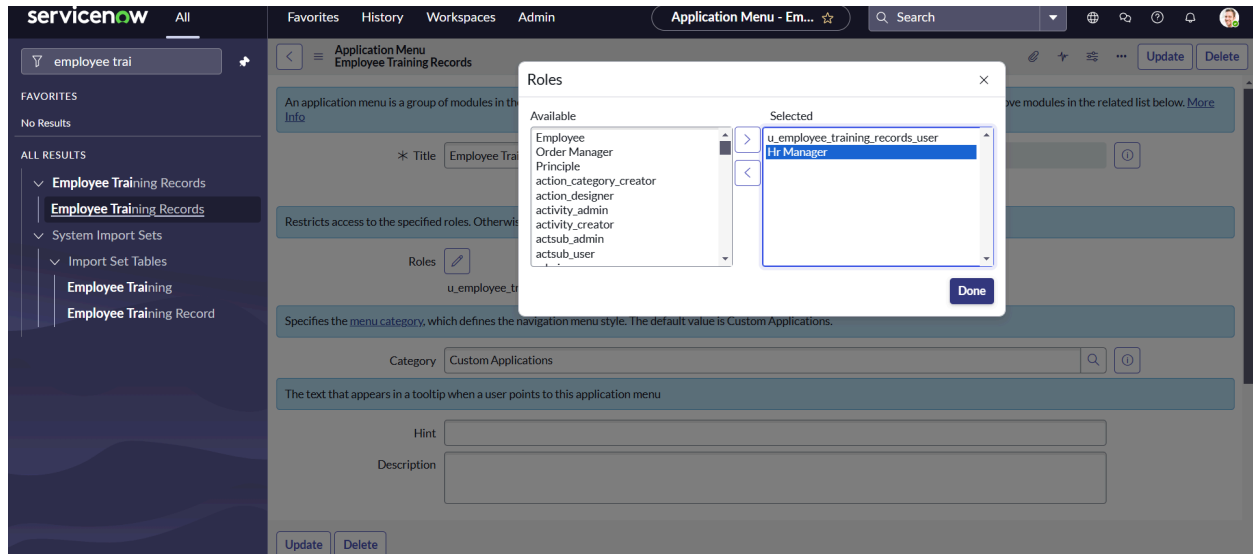


The screenshot shows the ServiceNow interface for the "backup user" form. The left sidebar contains the navigation menu with "Users" selected under "User Definition". The main form includes fields for "Web service access only", "Internal Integration User", and "backup user". Below the form are "Update", "Set Password", and "Delete" buttons. A section titled "Related Links" contains links for "View linked accounts", "View Subscriptions", and "Reset a password". A table titled "Entitled Custom Tables" shows the role "Hr Manager" assigned to the user "backup user".

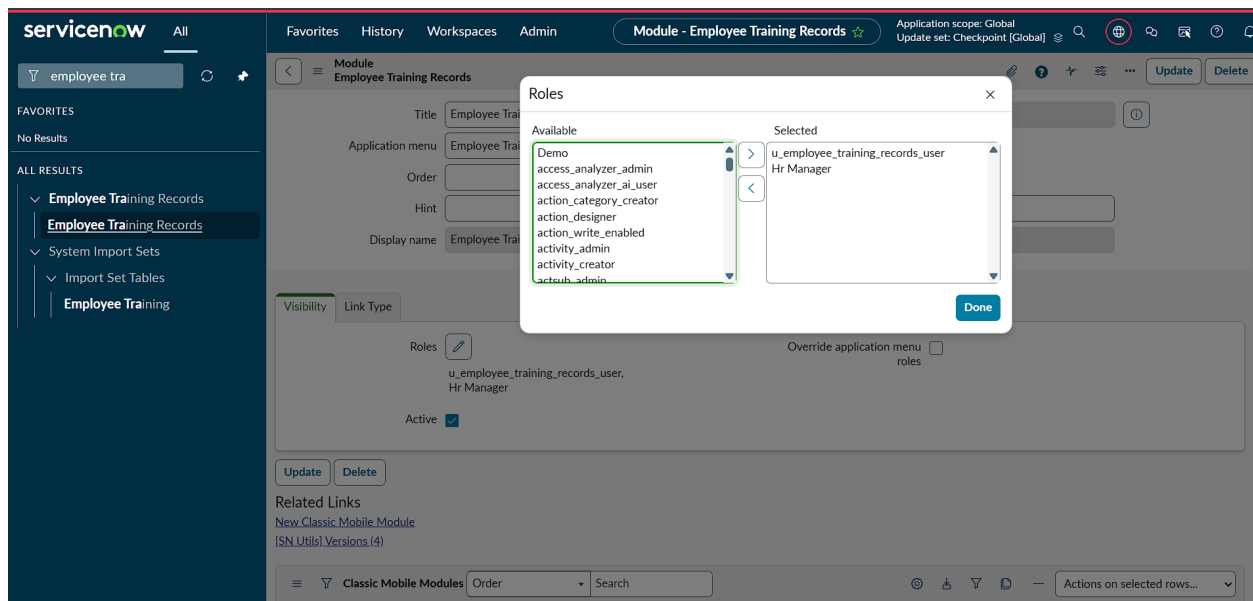
Role	State	Inherited	Inheritance Count
Hr Manager	Active	false	1

Activity 4: Add Roles to Applications menu and Modules

1. Open ServiceNow
2. Click All>>Employee Training Records (click the edit icon in the side).



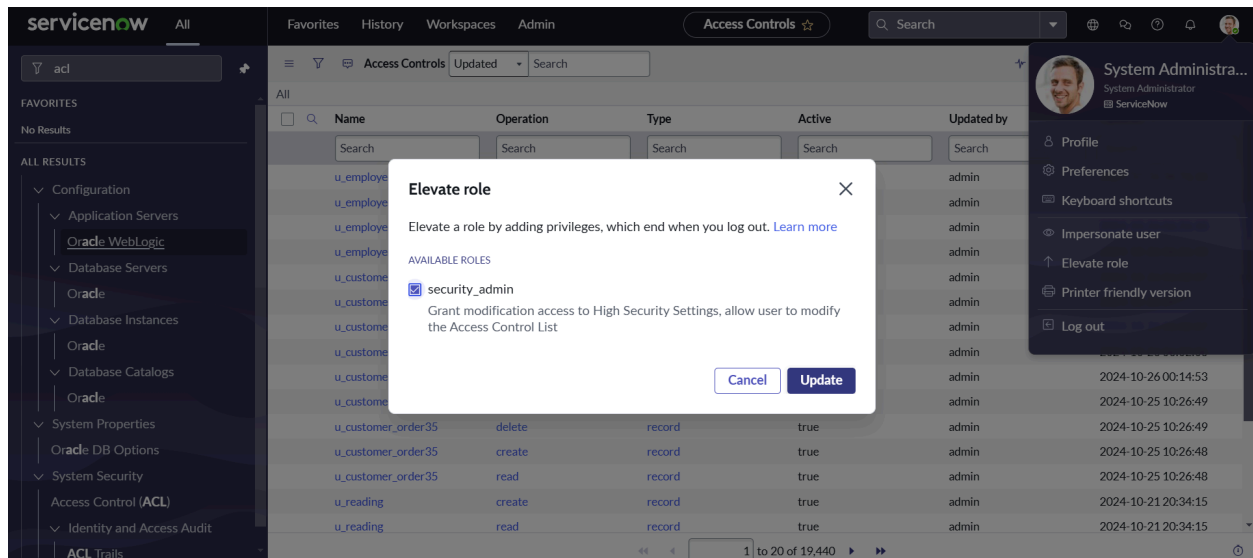
3. Click All>>Employee Training Records>>Employee Training Records (click the edit icon in the side).



Milestone 5 :Acl

Activity 1 : Update Role

1. Open ServiceNow
2. Click on Profile>>Elevate Role.
3. Check the Security_admin box and update.



The screenshot shows the ServiceNow interface with the 'Access Controls' tab selected. A search for 'acl' is performed. The 'Elevate role' dialog box is open, showing the 'security_admin' role selected. The dialog box text reads: 'Elevate a role by adding privileges, which end when you log out. [Learn more](#)'. Below this, it says 'AVAILABLE ROLES' and lists 'security_admin' with a description: 'Grant modification access to High Security Settings, allow user to modify the Access Control List'. There are 'Cancel' and 'Update' buttons at the bottom of the dialog box. The background shows a table of roles with columns: Name, Operation, Type, Active, and Updated by. The table lists various roles like 'u_employee', 'u_customer_order35', and 'u_reading'.

Activity 2:Acl creation

1. Create New ACL and give Read Access to Employee Training Records Table

servicenow All Favorites History Workspaces Admin Access Control - u_em... Search

Filter

Self-Service
Business Applications
Dashboards
Service Catalog
Employee Center
Knowledge
Visual Task Boards
Incidents
Watched Incidents
My Requests
Requested Items
Watched Requested Items
My Connected Apps
My Profile
My Tagged Documents
My Tags
My Knowledge Articles

Access Control - u_employee_training_records

* Type: record Application: Global

* Operation: read Active: ☒

Admin overrides: ☒ Advanced: ☐

Protection policy: -- None --

* Name: Employee Training Records [u_employee_training_records] -- None --

Description: Allow read for records in u_employee_training_records, for users with role Hr Manager.

Conditions

Access Control Rules allow access to the specified resource if all four of these checks evaluate to true:

1. The user has one of the roles specified in the **Role** list, or the list is empty.
2. Users that do not satisfy the Security Attributes or the Subject Conditions will be denied access to records matching this ACL.
3. Conditions in the **Condition** field evaluate to true, or conditions are empty.
4. The script in the **Script** field (advanced) evaluates to true, or sets the variable "answer" to true, or is empty.

The four checks are evaluated independently in the order displayed above.

[More Info](#)

Requires role

2. Give Hr Manager Role to the ACL

servicenow All Favorites History Workspaces Admin Access Control - u_em... Search

Filter

Self-Service
Business Applications
Dashboards
Service Catalog
Employee Center
Knowledge
Visual Task Boards
Incidents
Watched Incidents
My Requests
Requested Items
Watched Requested Items
My Connected Apps
My Profile
My Tagged Documents
My Tags
My Knowledge Articles

Access Control - u_employee_training_records

Description: Allow read for records in u_employee_training_records, for users with role Hr Manager.

Conditions

Access Control Rules allow access to the specified resource if all four of these checks evaluate to true:

1. The user has one of the roles specified in the **Role** list, or the list is empty.
2. Users that do not satisfy the Security Attributes or the Subject Conditions will be denied access to records matching this ACL.
3. Conditions in the **Condition** field evaluate to true, or conditions are empty.
4. The script in the **Script** field (advanced) evaluates to true, or sets the variable "answer" to true, or is empty.

The four checks are evaluated independently in the order displayed above.

[More Info](#)

Requires role

Role
☒ Hr Manager
☐ Insert a new row...

Security Attribute Condition

Local or Existing ☐ Existing ☒ Local

3. Create Another New ACL and Repeat the same Process to the Write Access

servicenow All Favorites History Workspaces Admin Access Control - u_em... Search

Filter

Self-Service
Business Applications
Dashboards
Service Catalog
Employee Center
Knowledge
Visual Task Boards
Incidents
Watched Incidents
My Requests
Requested Items
Watched Requested Items
My Connected Apps
My Profile
My Tagged Documents
My Tags
My Knowledge Articles

Access Control u_employee_training_records

* Type: record Application: Global

* Operation: write Active: ☒

Admin overrides: ☒ Advanced: ☐

Protection policy: -- None --

* Name: Employee Training Records (u_employee_training_records) -- None --

Description: Allow write for records in u_employee_training_records, for users with role Hr Manager.

Conditions

Access Control Rules allow access to the specified resource if all four of these checks evaluate to true:

1. The user has one of the roles specified in the **Role** list, or the list is empty.
2. Users that do not satisfy the Security Attributes or the Subject Conditions will be denied access to records matching this ACL.
3. Conditions in the **Condition** field evaluate to true, or conditions are empty.
4. The script in the **Script** field (advanced) evaluates to true, or sets the variable "answer" to true, or is empty.

The four checks are evaluated independently in the order displayed above.

[More Info](#)

Requires role

1 to 1 of 1

Milestone 6: Result

Activity 1: Testing Result

1. Impersonate the sys_user and Search Employee Training Records
2. Now You can see and edit the Fields

servicenow All Favorites History Employee Training Rec... Search

Employee Training Records Completion Date Search Actions on selected rows... New

Completion Date	Employee	Status	Training Name
2024-10-27	Abel Tuter	Inprogress	full stack.java
2024-12-31	Abel Tuter	Inprogress	Full Stack Java

servicenow All Favorites History Employee Training Rec... ☆

Employee Training Records
New record

Training Name Status -- None --

Completion Date

Department

[Submit](#)

3. Impersonate the other User you cannot See the Table

servicenow All Favorites History ServiceNow ☆

EMPLOYEE

FAVORITES
No Results

ALL RESULTS
Self-Service
Employee Center

Creator Studio

Create request-based apps quickly

A guided and curated environment for creating forms and assigning automations to them. No-code required.

[Open Creator Studio](#)

GO FURTHER

Power your workflow applications

Conclusion :

This project demonstrated the essential process of importing data into ServiceNow using Import Sets, leveraging dot-walking to access and utilize related table data efficiently, and applying Access Control Rules (ACLs) to enforce robust data security. By combining these core functionalities, we ensured accurate data integration, streamlined data relationships, and protected sensitive information through role-based access control.